

260330 C++ 作業

Ch9 Pointer

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Question 9

```
#include <iostream>
using namespace std;

void Cm_to_Inch(double *p)
{
    *p = (*p) * 0.394;
}

int main()
{
    cout << "請輸入欲轉換的公分數：";
    double x;  cin >> x;

    cout << x << " 公分 = ";
    Cm_to_Inch(&x);
    cout << x << " 英吋\n";
}
```

```
    return 0;  
}
```

9. 試撰寫一程式，輸入公分（cm）後轉換成英吋（inch），請利用指標傳入函數的方式完成。其轉換公式為 $1\text{ cm}=0.394\text{ inch}$ 。本題的執行結果範例如下所示：

請輸入欲轉換的公分數：**10**

10 公分=3.94 英吋

請輸入欲轉換的公分數：20

20 公分 = 7.88 英吋

Program finished with exit code 0

Press any key to exit

Question 14

```
#include <iostream>
using namespace std;

void square(int *arr)
{
    for (int i = 0; i < 5; ++i)
    {
        // 使用指標算術並解引用
        *(arr + i) = *(arr + i) * *(arr + i);
    }
}

int main()
{
    int arr[5] = {1, 2, 3, 4, 5};

    cout << "陣容內容為";
    for (int i = 0; i < 5; ++i) { cout << " " << arr[i]; }
    cout << "\n";

    square(arr);

    cout << "將每個元素平方，結果為";
    for (int i = 0; i < 5; ++i) { cout << " " << arr[i]; }
    cout << "\n";

    return 0;
}
```

14. 試撰寫一函數 `void square(int *arr)`，在呼叫 `square()` 函數後，一維陣列 `arr` 裡的每一個元素皆會被平方。請用指標的方式處理陣列。本題的執行結果範例如下所示：

陣列內容為 1 2 3 4 5

將每個元素平方，結果為 1 4 9 16 25

陣容內容為 1 2 3 4 5

將每個元素平方，結果為 1 4 9 16 25

Program finished with exit code 0

Press any key to exit



Question 17

```
#include <iostream>
#include <string>
#include <cctype>
using namespace std;
```

```
int main()
{
    cout << "Input source string: ";
    string s;
    getline(cin, s);

    cout << "Before process...\n";
    cout << "string = " << s << "\n\n";

    // 宣告一個字元指標 p，並指向字串 s 的第一個字元 (index 0)
    char *p = &s[0];
    while (*p != '\0')
    {
        *p = toupper(*p);
        p++;
    }

    cout << "After process...\n";
    cout << "string = " << s << "\n";

    return 0;
}
```


17. 試輸入一個字串，利用指標的方式將該字串的小寫字母轉換成大寫。本題的執行結果範例如下所示：

Input source string: *Out of sight, out of mind*

Before process...

string=Out of sight, out of mind

```
Input source string: Out of sight, out of mind
```

```
Before process...
```

```
string = Out of sight, out of mind
```

```
After process...
```

```
string = OUT OF SIGHT, OUT OF MIND
```

```
Program finished with exit code 0
```

```
Press any key to exit
```

Question 18

```
#include <iostream>
#include <string>
#include <cctype>
using namespace std;

int main()
{
    cout << "請輸入一個字串：";
    string s;
    getline(cin, s);

    // 宣告一個字元指標 p，並指向字串 s 的第一個字元 (index 0)
    char *p = &s[0];
    while (*p != '\\0')
    {
        if (*p == ' ') { *p = '*'; }
        p++;
    }

    cout << "替換後的字串 = " << s << "\\n";

    return 0;
}
```

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試由鍵盤輸入一字串，利用指標的方式將該字串中所有的空白以星號*代替。本題的執行結果範例如下所示：

請輸入一個字串：*Out of sight, out of mind*

替換後的字串=*Out*of*sight,*out*of*mind*

```
請輸入一個字串：Out of sight, out of mind  
替換後的字串 = Out*of*sight,*out*of*mind
```

```
Program finished with exit code 0  
Press any key to exit
```

